

INDIAN INSTITUTE OF TECHNOLOGY GUWAHATI

ESTATES AND WORKS SECTION • ROADS & CIVIL INFRASTRUCTURE DIVISION
Predictive Pavement Management System — Official Work Order

Work Order ID: WO-IITG-20260925-01	Date Issued: 25 September 2026, 20:41 IST
Target Route: Core V North Loop	Action Type: Full Segment Resurfacing
Center GPS: 26.19105, 91.69243	Defect Cluster Count: 15 detected defects

1. PREDICTIVE ENGINEERING & RISK ASSESSMENT

This 100-meter corridor was clustered via Spatial DBSCAN combining multi-modal inputs (proactive CCTV traffic telemetry, 100 Hz bicycle accelerometer vertical shocks & crowdsourced reports). The aggregate **Road Damage Index (RDI) is 0.372** with a cumulative surface defect area of **24.21 m²**. Compounded by pavement aging, the **Predictive Degradation Index (PDI) has reached 0.927**, crossing the critical threshold for sub-base deterioration. Reactive point patching is hereby discontinued for this segment in favor of predictive 100m cold-milling and asphaltic concrete resurfacing.

2. LIFECYCLE COST SAVINGS ANALYSIS (GRANICA COMPLIANT)

Metric / Parameter	Reactive Spot Patching	Predictive Full Resurfacing
Immediate Repair Cost	Rs. 112,500.00	Rs. 34,000.00
Expected Longevity	6 - 9 Months (High monsoon failure)	4 - 5 Years (Durable wear course)
Projected 3-Year Maintenance	Rs. 281,250.00	Rs. 34,000.00
Net Campus Savings	Baseline (0% Savings)	68.0% Cost Reduction

3. SEGMENT DEFECT MANIFEST

Observation ID	Source	Area (m²)	Peak Shock	PDI Score
OBS_IITG_0001	CCTV	3.53	3.52g	2.037
OBS_IITG_0009	CCTV	2.48	1.59g	1.349
OBS_IITG_0017	STUDENT_APP	1.79	1.00g	0.726
OBS_IITG_0025	CCTV	1.90	1.33g	0.872
OBS_IITG_0033	CCTV	2.23	1.61g	0.933
OBS_IITG_0041	STUDENT_APP	0.52	1.00g	0.410

4. AUTHORIZATION & SIGN-OFF

Recommended By:

Lead Data Scientist / AI Vision Engine
IIT Guwahati Estates Automation

Approved By:

Executive Engineer (Civil)
Estates & Works Section, IIT Guwahati